

CardioInsight

Heart Disease Prediction & Clinical Risk Analysis

End-to-end Machine Learning workflow: EDA → preprocessing → model comparison → SVM selection → deployment

NTI Machine Learning Training

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Project Overview

01

- Predict the presence of heart disease from demographic, lifestyle, and clinical indicators.
- Perform exploratory data analysis and identify relationships between features and the target.
- Build a reproducible preprocessing pipeline using encoding, feature engineering, scaling, and stratified splitting.
- Train and compare 9 classification models using Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
- Select and serialize the SVM (RBF) model for the CardioInsight application.

50,000 original records

21 original columns

**22 columns after
engineering**

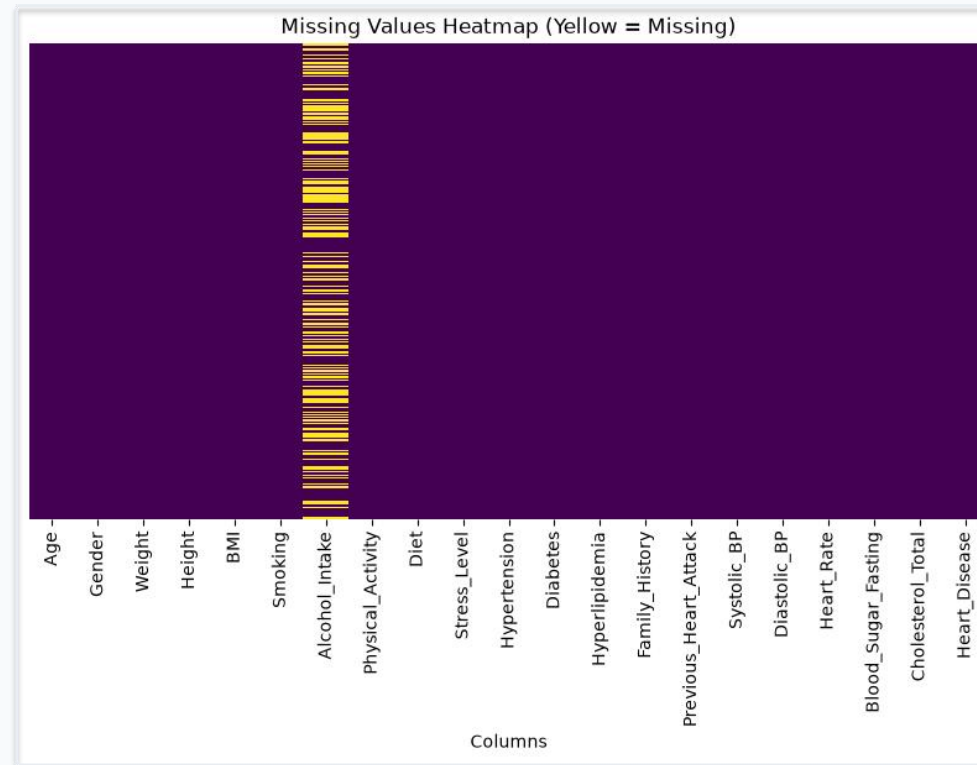
Initial Dataset Inspection

02

Shape: 50,000 rows × 21 columns

Duplicates: 0

Missing values: Alcohol_Intake = 20,109

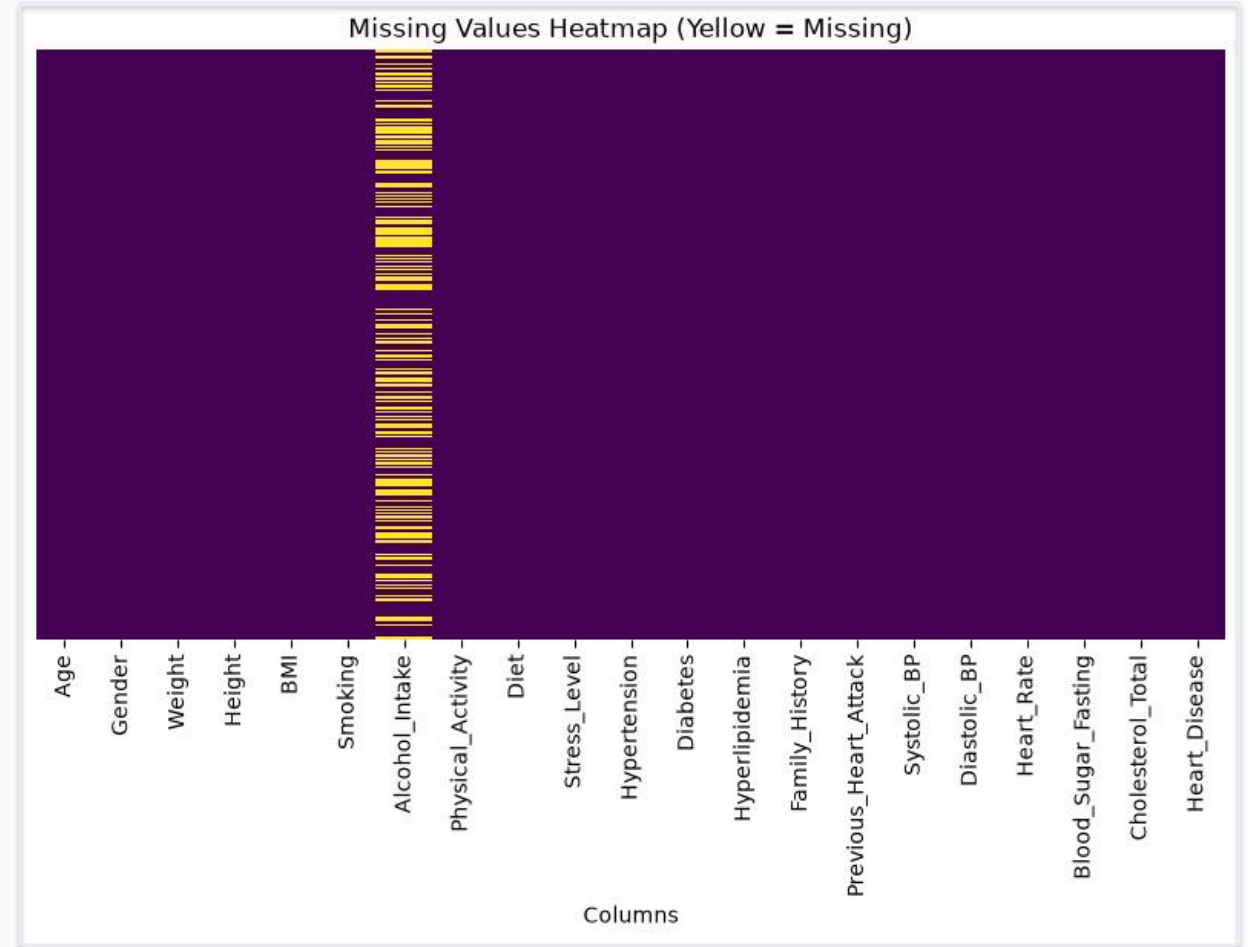


Target variable: Heart_Disease (0 = No, 1 = Yes)

Missing Values & Data Quality

02

- Only Alcohol_Intake contains missing values in the original dataset.
- 20,109 missing entries were identified.
- The preprocessing step fills missing Alcohol_Intake values with 'None'.
- Rows with invalid blood-pressure ordering (Systolic_BP $\leq$ Diastolic_BP) are removed.



Missing Values Heatmap

Feature Analysis

02

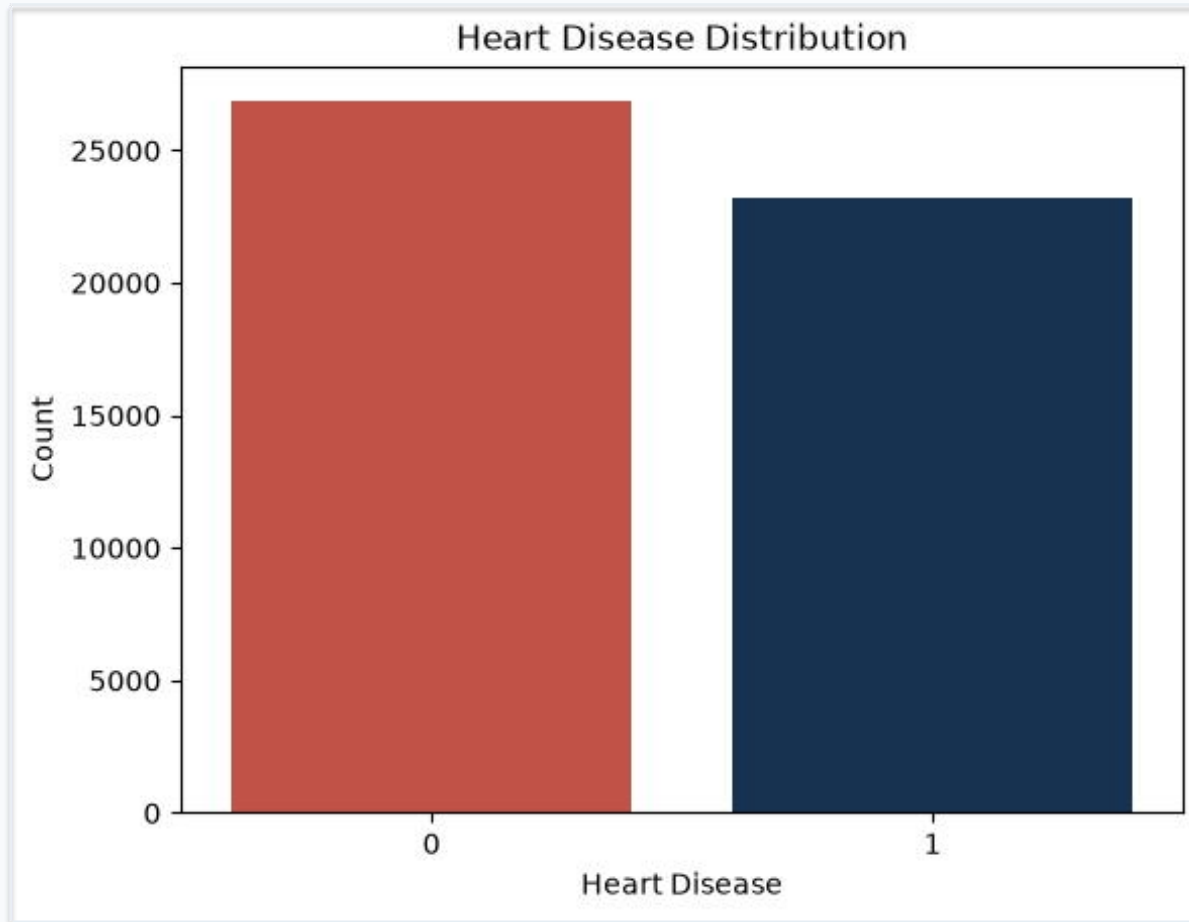
Feature	Unique Count	Values / Range
Age	50	30–79
Gender	2	Male / Female
Weight	70	50–119
Height	50	150–199
BMI	221	18.0–40.0
Smoking	3	Never / Current / Former
Alcohol_Intake	3	None / Low / Moderate / High*
Physical_Activity	3	Sedentary / Active / Moderate
Diet	3	Healthy / Average / Unhealthy
Stress_Level	3	Low / Medium / High
Heart_Disease	2	0 / 1

- Numerical features were summarized with descriptive statistics.
- Categorical features were inspected using frequency and unique-value analysis.
- * Alcohol_Intake initially contains missing values before cleaning.

Feature inspection supports the preprocessing design

Target Distribution

02

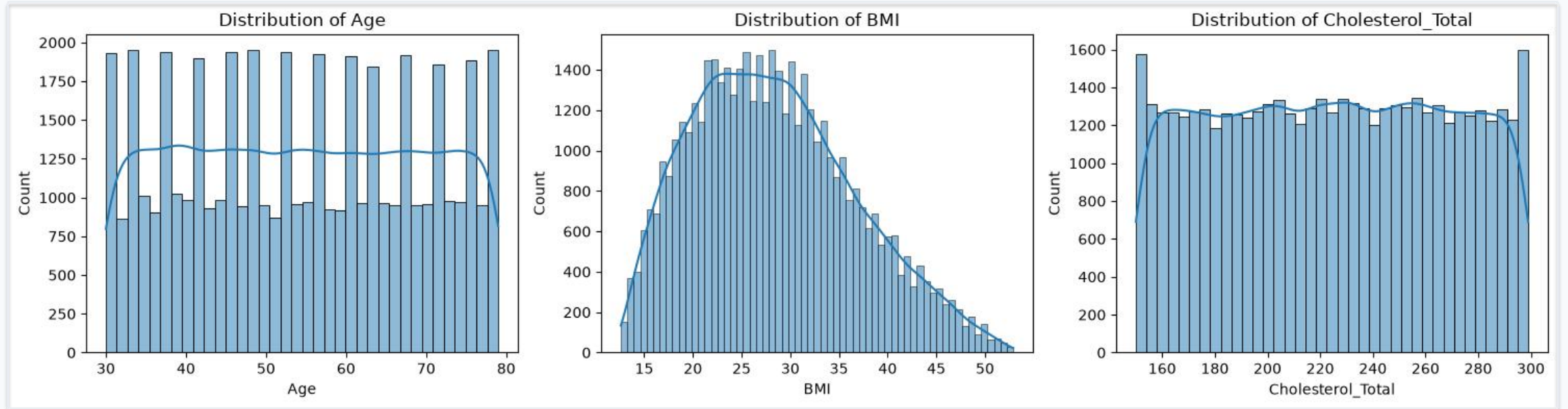


Class	Count	Percentage
0 — No Heart Disease	26,827	53.65%
1 — Heart Disease	23,173	46.35%

- The two classes are relatively balanced.
- No heavy resampling strategy was used in the notebook.
- The target is suitable for binary classification.

Numerical Feature Distributions

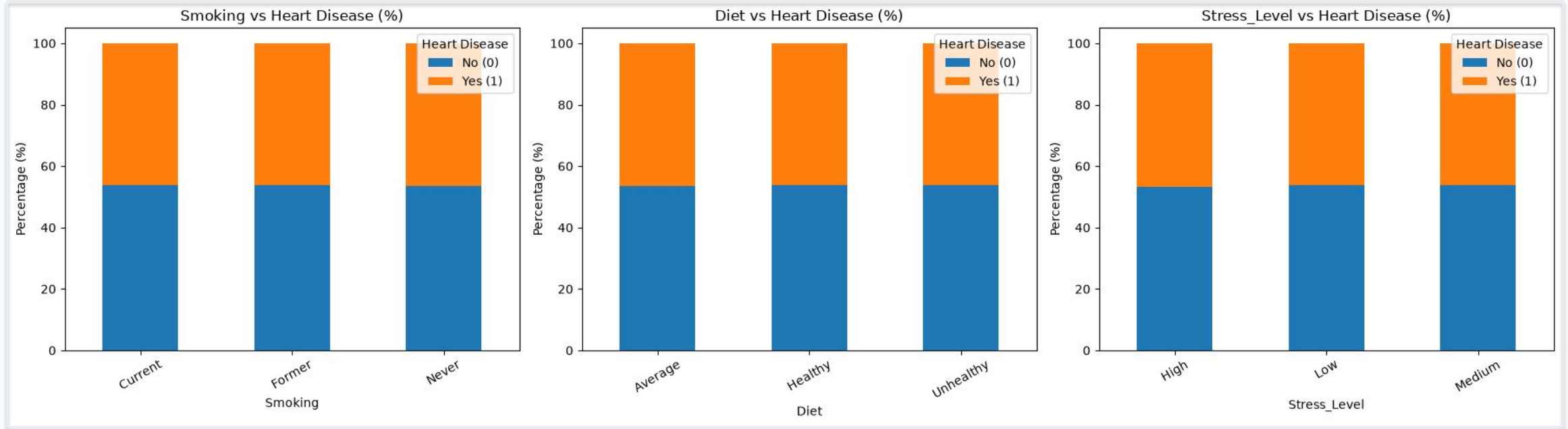
03



- Age is broadly distributed across the available age range.
- BMI is concentrated around the middle of its range.
- Cholesterol_Total spans a wide range and shows a relatively broad distribution.

Categorical Features vs. Heart Disease

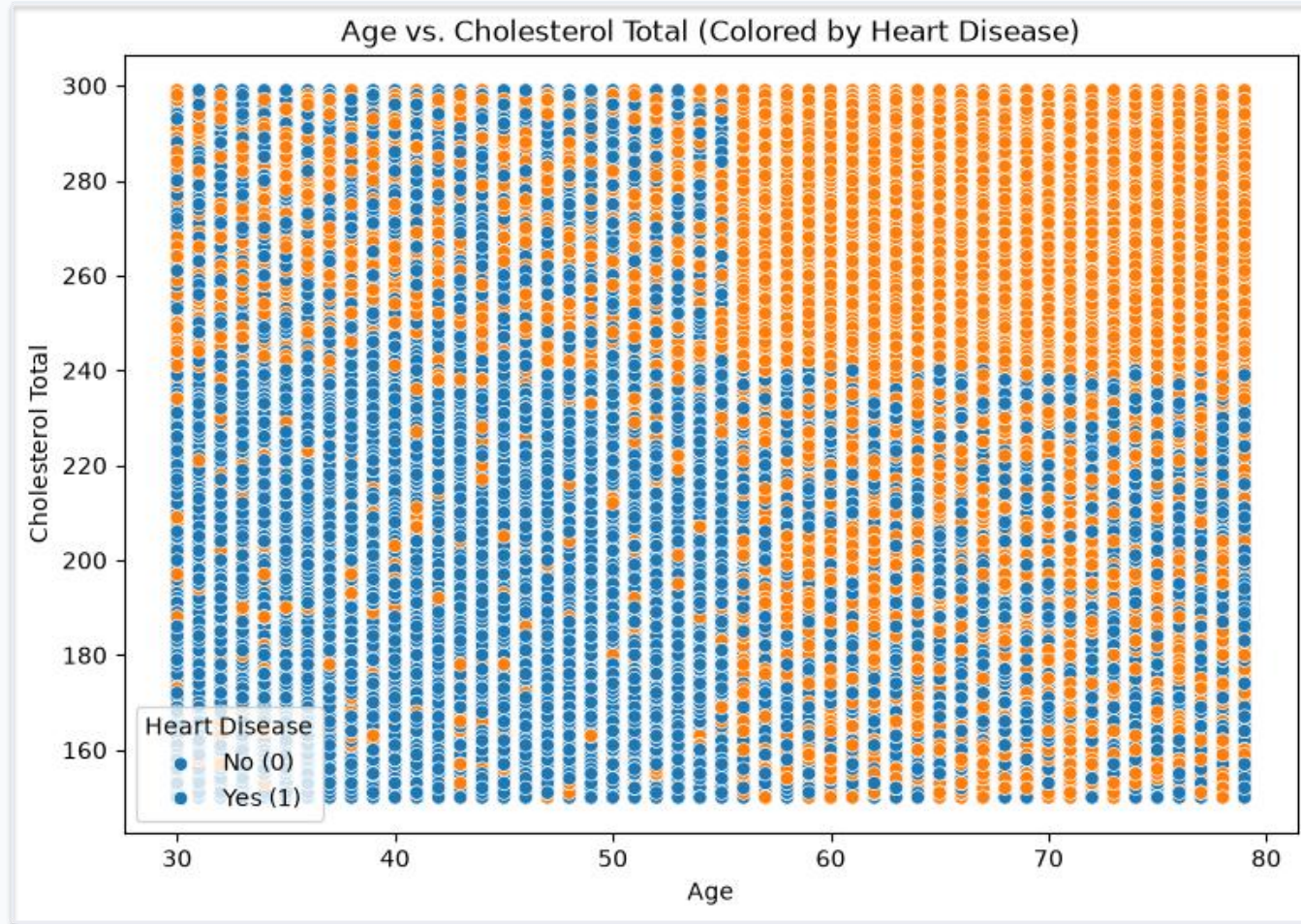
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- Smoking, Diet, and Stress_Level show similar class proportions across their categories.
- These features do not provide strong standalone separation in the plotted distributions.

Age vs. Total Cholesterol

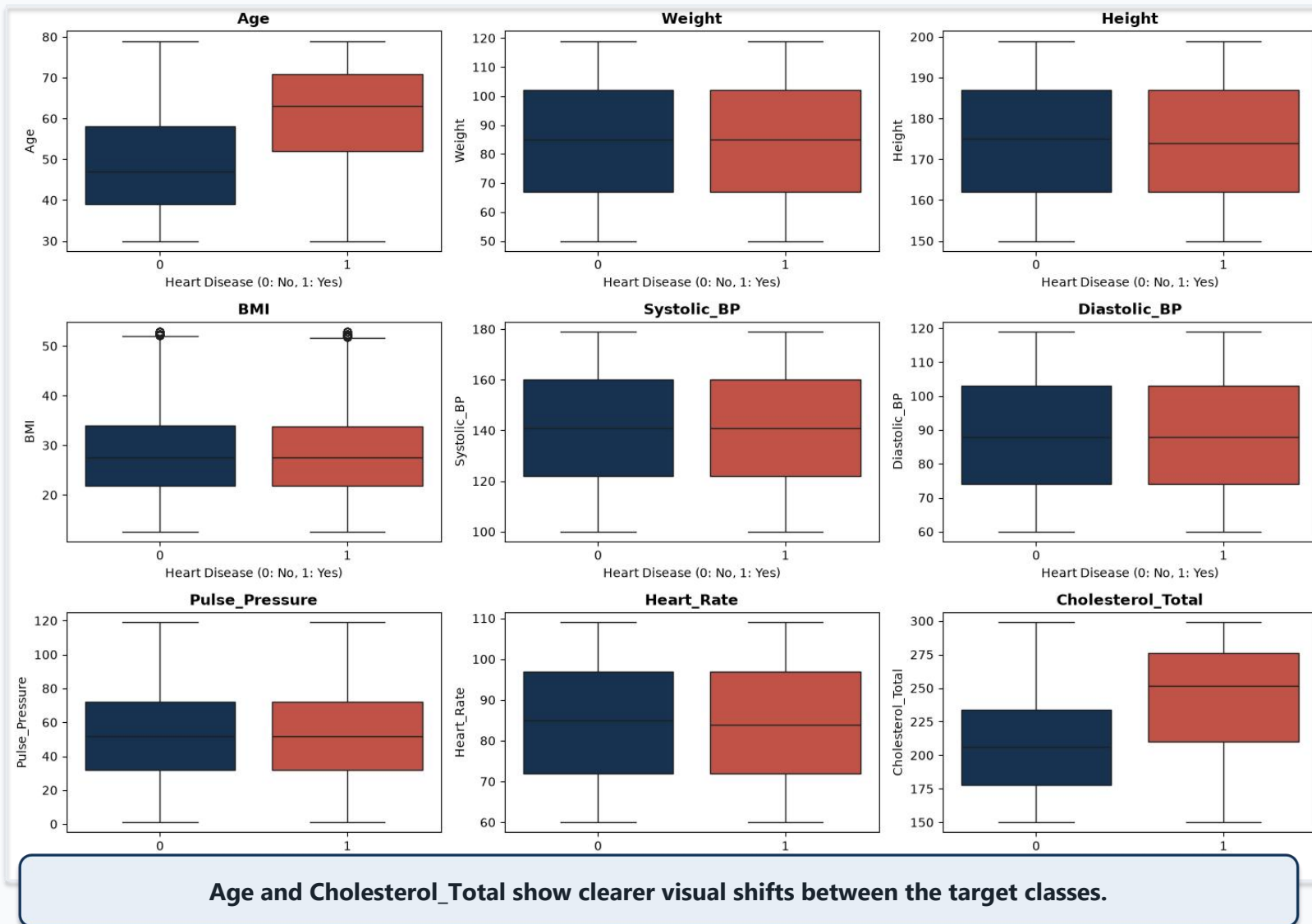
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- The two target classes are heavily interleaved.
- Age and Cholesterol_Total alone do not create a clear linear decision boundary.
- This supports testing models capable of capturing multivariate and non-linear patterns.

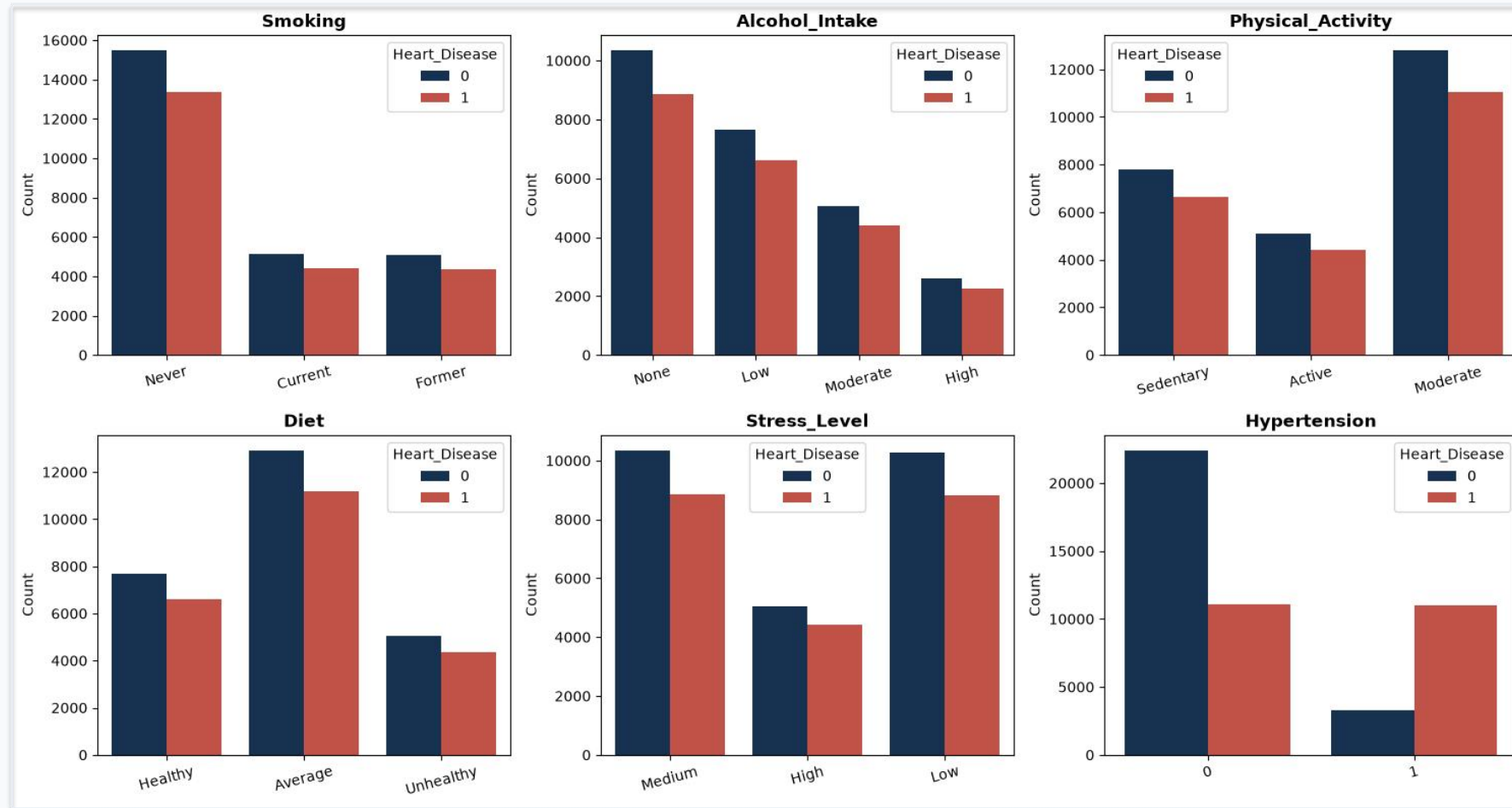
Numerical Features: Boxplots

03



Lifestyle & Clinical Categorical Plots

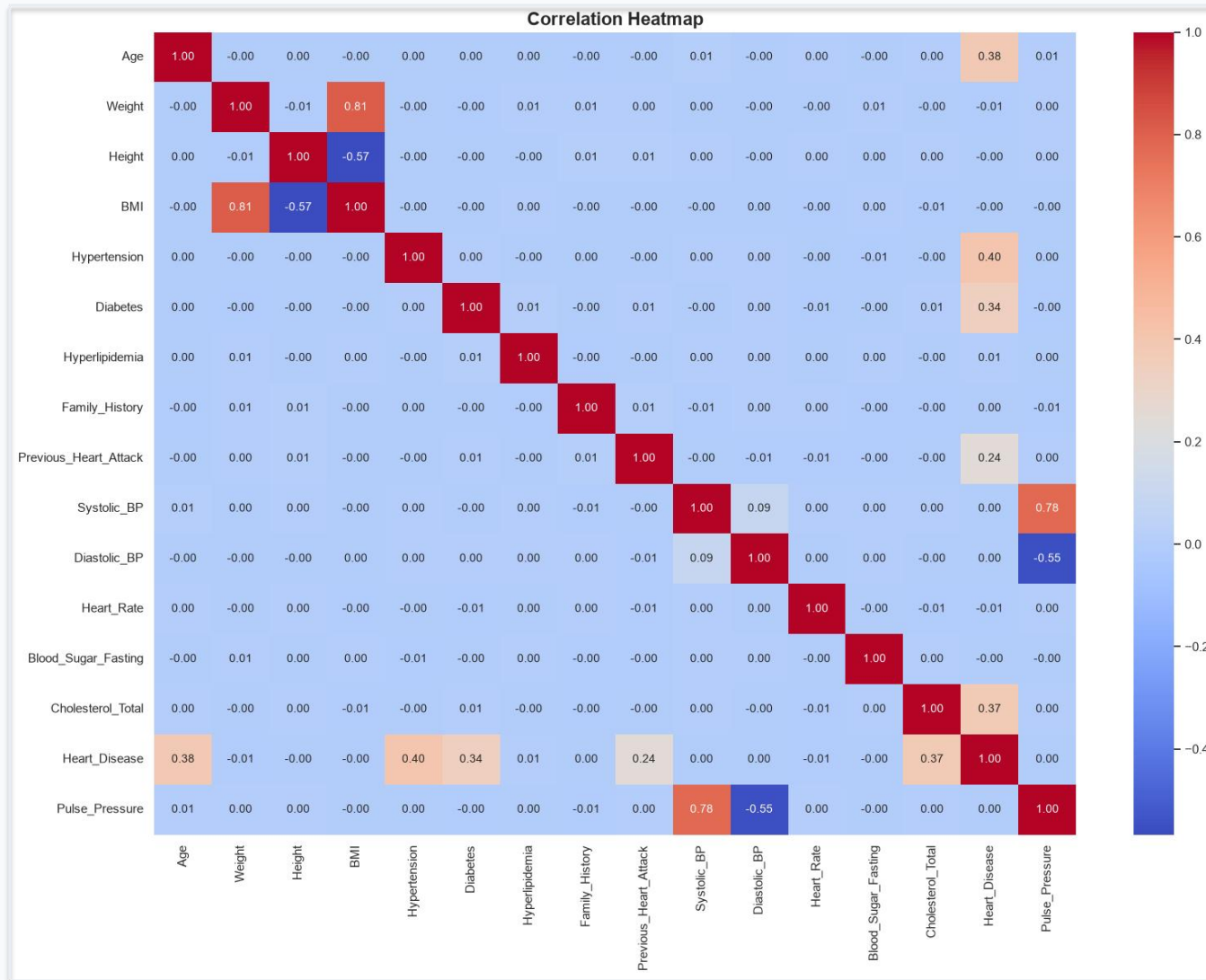
03



Hypertension shows a noticeably stronger contrast between the two target classes.

Correlation Heatmap

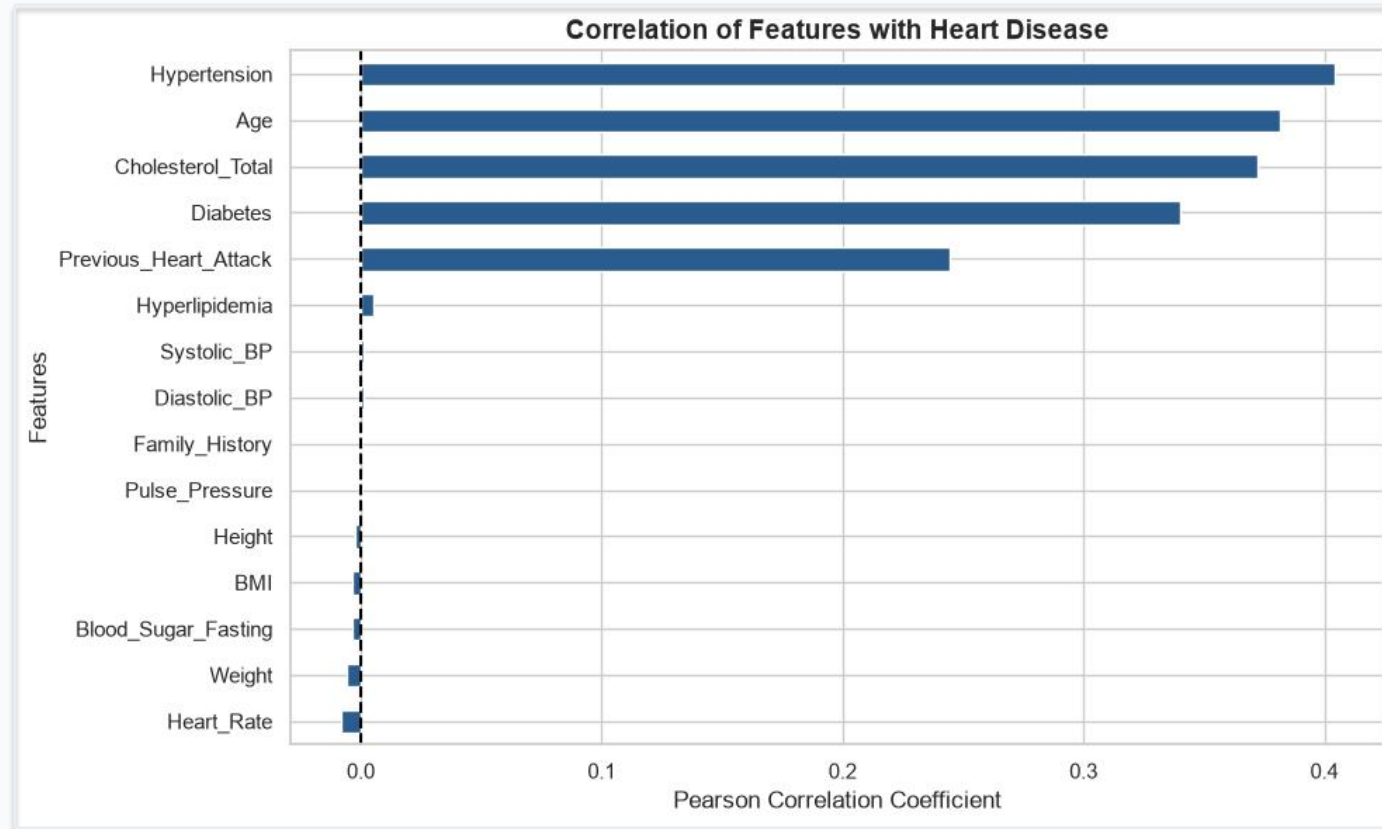
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- Heart_Disease shows moderate positive correlations with Hypertension (0.40), Age (0.38), Cholesterol_Total (0.37), and Diabetes (0.34).
- Weight and BMI are strongly correlated (0.81).
- Systolic_BP and Pulse_Pressure are strongly correlated (0.78).

Feature Correlation with Heart Disease

03



- Strongest positive correlations: Hypertension, Age, Cholesterol_Total.
- Moderate positive associations: Diabetes and Previous_Heart_Attack.
- Several features have weak or near-zero linear correlation with the target.

Data Cleaning & Feature Engineering

04

- Alcohol_Intake missing values → filled with 'None'.
- Invalid blood-pressure rows → keep only Systolic_BP > Diastolic_BP.
- BMI recomputed from Weight and Height.
- Pulse_Pressure engineered as Systolic_BP – Diastolic_BP.
- Cleaned dataset shape: 47,765 rows × 22 columns.

Original

50,000 × 21

Cleaned

47,765 × 22

Engineered Features

- BMI
- Pulse_Pressure

Binary Classification

21 input features before preprocessing

80% Train / 20% Test

- Target: Heart_Disease
- Train/Test split uses random_state=42 and stratify=y.
- Training set: 38,212 rows × 21 features.
- Testing set: 9,553 rows × 21 features.

- Ordinal Encoding → Diet, Stress_Level, Physical_Activity
- One-Hot Encoding → Gender, Smoking, Alcohol_Intake
- Column alignment ensures train/test feature structures match.
- StandardScaler applied to numerical features for scale-sensitive models.
- Final scaled training shape: 38,212 × 23.

Model	Test Accuracy
Logistic Regression	92.52%
KNN	83.22%
Naive Bayes	83.59%
SVM (RBF)	96.20%
Decision Tree	95.89%
Random Forest	99.82%
Gradient Boosting	100.00%
AdaBoost	100.00%
XGBoost	99.54%

- Nine classifiers were trained and evaluated using the same target and split.
- Tree-based ensembles reached extremely high test scores on the synthetic dataset.
- SVM (RBF) achieved 96.20% test accuracy with a 0.9946 ROC-AUC.

All models trained and evaluated successfully

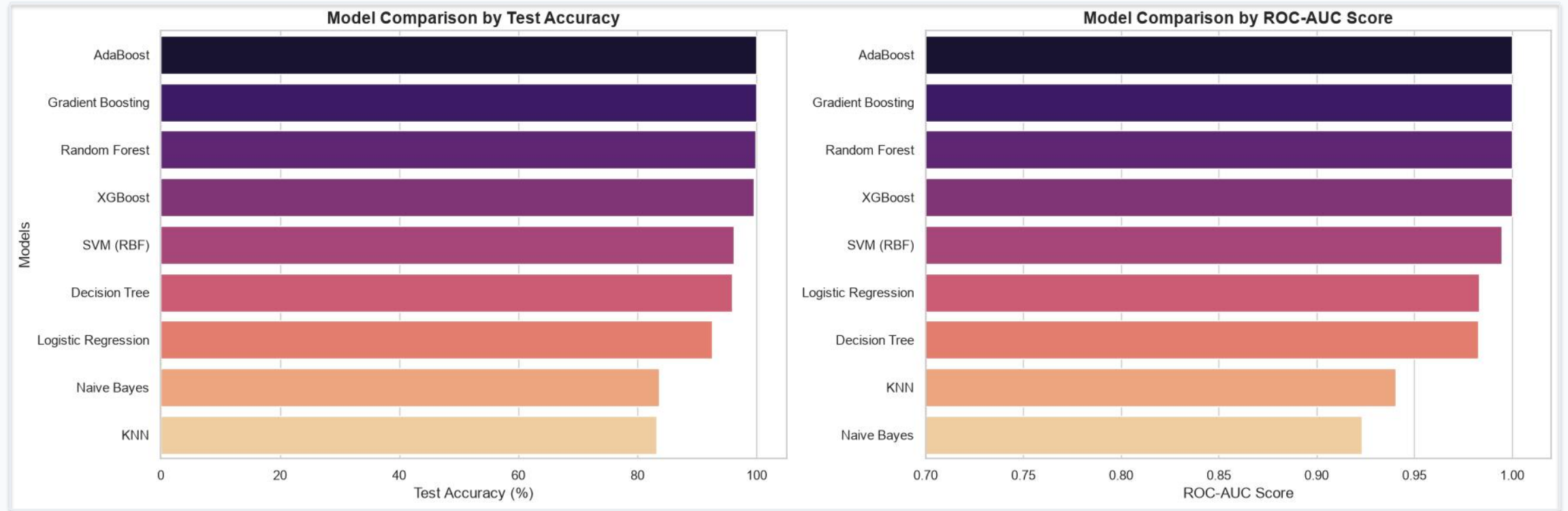
Final Model Comparison

05

Model	Train Acc.	Test Acc.	ROC-AUC	Recall	Precision	F1
AdaBoost	100.00	100.00	1.0000	1.0000	1.0000	1.0000
Gradient Boosting	100.00	100.00	1.0000	1.0000	1.0000	1.0000
Random Forest	99.83	99.82	1.0000	0.9962	1.0000	0.9981
XGBoost	99.53	99.54	1.0000	0.9900	1.0000	0.9950
SVM (RBF)	98.63	96.20	0.9946	0.9604	0.9576	0.9590
Decision Tree	95.89	95.89	0.9828	0.9111	1.0000	0.9535
Logistic Regression	92.49	92.52	0.9830	0.9231	0.9158	0.9195
Naive Bayes	83.45	83.59	0.9232	0.7935	0.8427	0.8173
KNN	85.38	83.22	0.9405	0.6879	0.9317	0.7914

Model Comparison Charts

05



- The charts show the gap between raw test accuracy and ROC-AUC across the nine models.
- The SVM combines high discrimination with a lower test score than the synthetic-data tree ensembles.

Selected Model: SVM (RBF)

06

SVM (RBF)

Metric	Result
Train Accuracy	98.63%
Test Accuracy	96.20%
ROC-AUC	0.9946
Recall	0.9604
Precision	0.9576
F1-Score	0.9590

- SVC with RBF kernel, C=10, probability=True.
- The selected model preserves strong class-separation performance while avoiding the perfect test scores observed in several tree-based models.
- The choice is documented as the project's preferred model for CardioInsight.

Evaluation & Clinical Deployment Analysis

06

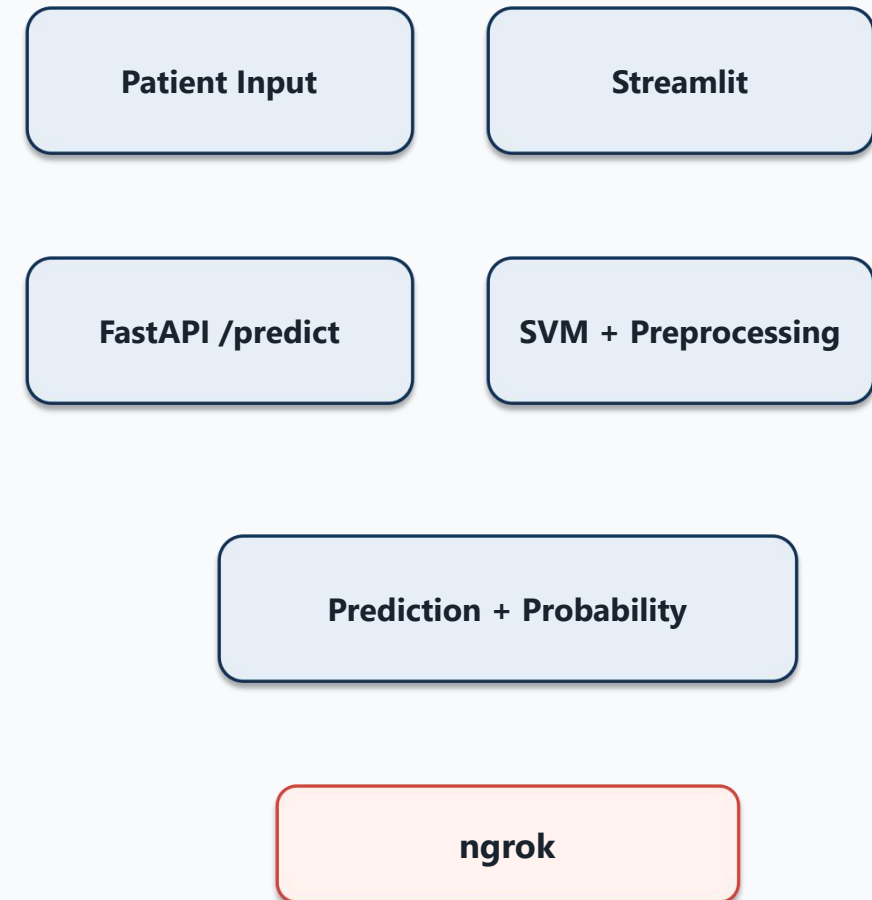
- The notebook flags 100% accuracy on synthetic data as a potential generalization concern.
- Because the dataset is synthetic and generated from clean rules, extremely high tree-model scores may reflect the structure of the generator.
- False negatives are important to monitor in a heart-disease screening context.
- Threshold tuning to 0.35 is discussed in the notebook analysis, but the supplied deployment code currently uses the model's default `predict()` decision.
- Any threshold change should be evaluated on validation/test data using Recall, Precision, F1, and a confusion matrix.

Deployment should be treated as decision support, not a medical diagnosis.

Model Serialization & Deployment

07

- The trained SVM is saved as `best_model.pkl`.
- The saved bundle contains the model, `StandardScaler`, `OrdinalEncoder`, and training feature columns.
- FastAPI receives patient data, reproduces the preprocessing steps, and returns prediction + probability.
- Streamlit provides the user interface and sends requests to the FastAPI endpoint.
- Google Colab + ngrok can expose the API and Streamlit interface for a temporary live demo.



Live Demo: CardioInsight Interface

07

The screenshot shows a web browser window with the URL `runaround-deed-demanding.ngrok-free.dev`. The interface is divided into two main sections: **Personal Information** and **Medical Information**. The **Personal Information** section includes fields for Age (50), Gender (Male), Weight (kg) (70.00), Height (cm) (170.00), Smoking (Never), Alcohol Intake (None), and Physical Activity (Sedentary). The **Medical Information** section includes fields for Hypertension (0), Diabetes (0), Hyperlipidemia (0), Family History (0), Previous Heart Attack (0), Systolic BP (120.00), and Diastolic BP (0). A blue button labeled "Allow" is visible in the top right corner of the browser window. A light blue box at the bottom of the interface contains the text "Patient information + clinical measurements".

Personal Information	Medical Information
Age: 50	Hypertension: 0
Gender: Male	Diabetes: 0
Weight (kg): 70.00	Hyperlipidemia: 0
Height (cm): 170.00	Family History: 0
Smoking: Never	Previous Heart Attack: 0
Alcohol Intake: None	Systolic BP: 120.00
Physical Activity: Sedentary	Diastolic BP: 0

Patient information + clinical measurements

Live Demo: Prediction Result

07

The screenshot shows a web browser window with multiple tabs. The active tab is titled 'CardioInsight' and the address bar shows 'runaround-deed-demanding.ngrok-free.dev'. A notification bar at the top of the browser says 'Begin browsing instantly: Chrome can now launch when Windows starts.' with an 'Allow' button. The web application interface is dark-themed and contains the following elements:

- A dropdown menu at the top left with 'LOW' selected.
- A 'Total Cholesterol' input field with the value '200.00' and minus/plus buttons.
- A 'BMI' label with the value '24.2'.
- A 'Pulse Pressure' label with the value '40.0'.
- A 'Predict Heart Disease' button.
- A green banner displaying the result: 'No Heart Disease Detected'.
- A 'Prediction Probability' label with the value '0.00%'.

Example output: No Heart Disease Detected • Prediction Probability: 0.00%

Key Takeaways

- EDA identified important patterns and dataset structure before modeling.
- Cleaning and feature engineering produced a consistent $47,765 \times 22$ dataset.
- Nine models were compared using multiple classification metrics.
- SVM (RBF) was selected with 96.20% test accuracy and 0.9946 ROC-AUC.
- CardioInsight connects the trained model to a FastAPI backend and Streamlit interface.

Thank You